

Solution Requirements

Date	04 July 2026
TeamID	
ProjectName	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely log in using a username and password to access the HDI prediction system.	High
2	Authorization levels	Admin can manage the dataset and model, while users can enter country indicators and view prediction results.	High
3	External interfaces	The system connects to the trained Machine Learning model and dataset for generating HDI predictions.	High

4	Transactions processing	The system accepts user input (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI per Capita), processes the data, and predicts the HDI category.	High
5	Reporting	Displays the predicted HDI category and allows users to view or download prediction results	Medium
6	Businessrules, etc.	The prediction is based only on valid numerical inputs within the accepted range. The trained ML model classifies the country into Very High, High, Medium, or Low HDI.	High
7	Compliance to laws or regulations	User data is processed securely, and the application follows general data privacy and ethical AI practices.	Medium
8	<i>Other</i>	The system provides a simple, user-friendly interface for entering indicators and viewing prediction results instantly	Medium

Step2:Non-FunctionalRequirements(NFR)

S.No	NFRCategory	RequirementDescription	Target Metric / AcceptanceCriteria
1	Performance& Speed	The system should generate HDI predictions quickly after user input.	Prediction completed within 2 seconds
2	Scalability	The application should support multiple users making predictions simultaneously.	Supports 500+ concurrent users without performance degradation.
3	Security&Data Privacy	User inputs should be protected, and communication should be secure.	Secure login with encrypted communication (HTTPS).
4	Reliability& Availability	The system should remain available with minimal downtime.	99.9% uptime per month
5	Usability &Accessibility	The application should provide an intuitive interface that is easy for users to understand and operate.	Responsive design with simple navigation and browser compatibility.
6	<i>Other</i>	The application should be maintainable and allow easy model updates when new HDI data becomes available.	Model and dataset can be updated without affecting system functionality.